

4.3.3.1.4 FE Analysis

Firstly, a rough 3D model of the upright respecting all the constraints was created, in order to undergo topological optimization. This provided information on a possible distribution of the material, which had to be refined and adapted to the possibilities of CNC machinery. The designing process continued on the 3D model, in order to find the most suitable geometries for the aforementioned goals.

The first FE analysis have been run on the single component, in order to evaluate the stresses of the structure. A simplified carrier cover has been added to the component in order to obtain a better distribution of the torque on the geartrain. Secondly, a model with the whole upright assembly was prepared, with model loads and boundary conditions are applied to the upright through the contact of all the elements attached (bolts, mountings, bearings) and so through pressure and shear stresses.

Part	Functions	Material properties	Mesh
Upright	Main component to be analysed	Aluminium $E=70000 \text{ MPa}$ $\nu=0.3$	TET (quadratic)
Upper whisbone terminal	It connects the upright to the upper A-Arm through two bolts	Aluminium	TET (quadratic)
Bolts	They connect the mounting, the caliper and the tie rod to the upright, as well as the mounting to the upper a-arm.	Steel $E=210000 \text{ MPa}$ $\nu=0.3$	HEX (linear)
Spheres	Part of the joint, allow a rotational degree of freedom to the a-arm.	Steel	HEX (linear)
Bushings	They distribute the preload of the bolts on a bigger surface	Steel	HEX (linear)
Dummy Geartrain	It transfers the torque to the upright	Steel	HEX (linear)
Dummy Hub	It simulates the brake disc and the hub with the races of the bearings	Aluminium (hub) Steel (internal races of bearings) Titanium	HEX (linear)

		E= 110000 MPa (brake disc) $\nu=0.3$	
Dummy Caliper	It simulates the brake caliper and the two bolts connecting it to the upright	Aluminium (caliper) Steel (bolts)	HEX (linear)
Seal protection ring	Sealing element for lubricants	Steel	HEX (linear)

Table 4-10 - properties

	Front Upright	Mounting	Bolts	Annulus Gear	Hub	Caliper
Boundary conditions	UBJ: hinged LBJ: X, Y translations locked S_U: Y-translation locked	Position locked by bolts	-	Position locked by bearings and hub	Disc tied to the brake pads	Position locked by bolts
Loads- Maximum Braking:	-3750 N (x) ground +270 (y) ground +2400 N (z) ground 300 Nm(y), distributed on the 2 stages of the dummy geartrain	-	13 kN preload (mounting) 10 kN preload (lbj)		2 kN preload on the hub	8 kN preload (bolts)

Interaction properties

Surface to surface hard contact

Friction 0.3	Frictionless	Tie
Upright – Mounting	Dummy hub – Upright	Brake pad – Brake disk
Upright – Bolts		
Upright – Bearing races		

Upright - seal protection ring

Mounting - Bolts

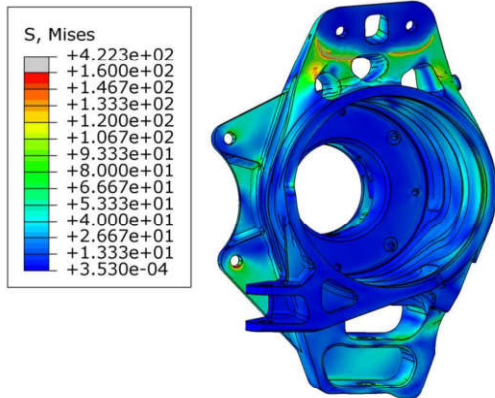


Figure 4-43: Front left upright max braking load inside view, scale factor 10

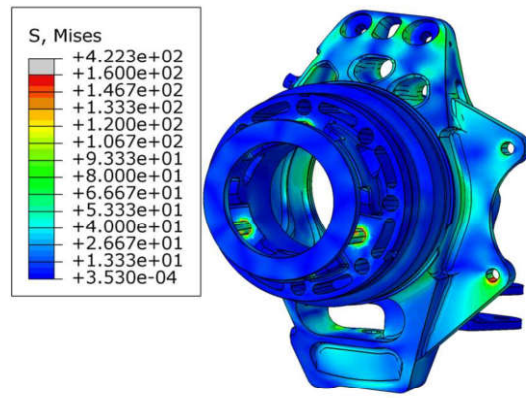


Figure 4-42: Front left upright max braking load outside view, scale factor 10

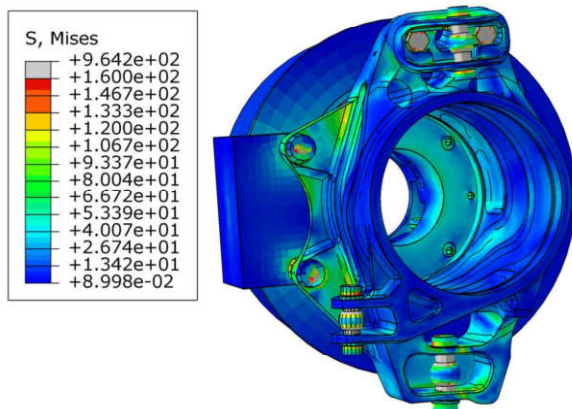


Figure 4-40: Front left upright max braking, inside view with hub, caliper and bolts, scale factor 10

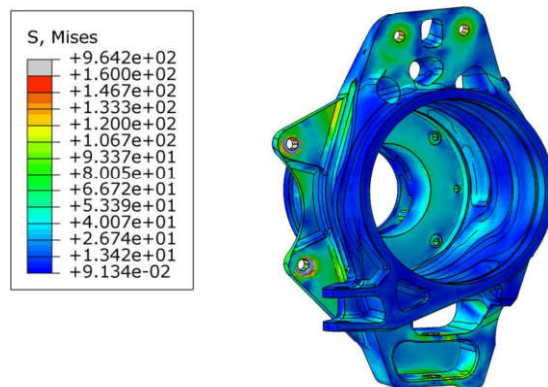


Figure 4-41: Front left upright max braking, inside view, scale factor 10

4.3.4 Rear Uprights

4.3.4.1 Design process

4.3.4.1.1 Load Analysis

The loads are provided by the Vehicle Dynamics department. There are different possible load cases, the most critical one being the maximum braking. The loads considered are the forces exchanged with both the ground and the brake caliper, and the torque due to the acceleration or regenerative braking on the y axis.

4.3.4.1.2 Material choice

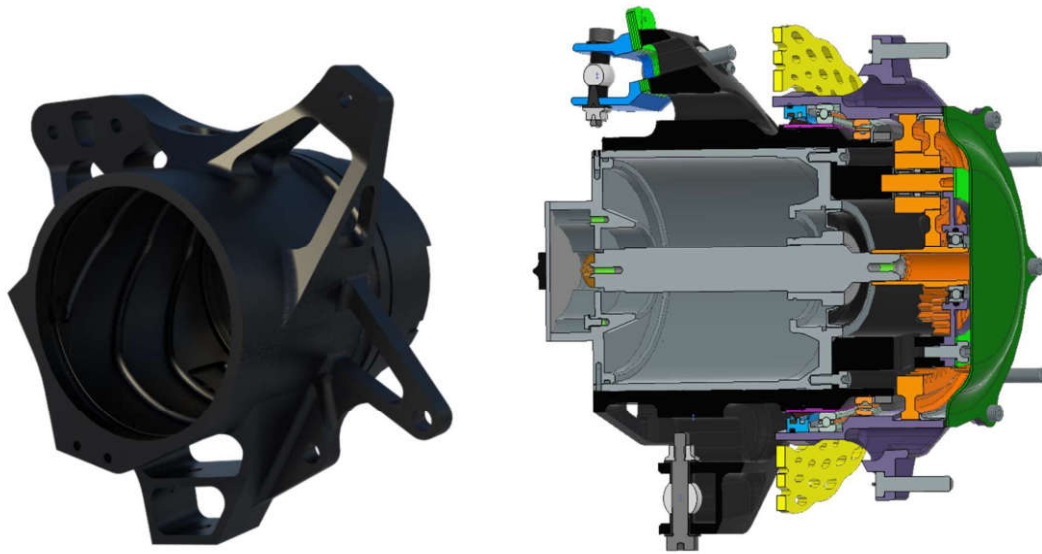
The material parameter to be optimized is $\frac{\sigma_f^{2/3}}{\rho}$, therefore the chosen material is the 7075-T6 alloy, which provides a good combination of lightness, stiffness and manufacturability. RPO.2 is 390 MPa.

4.3.4.1.3 3D Model

The model consists of:

- A cylindrical case, which were sized by the drivetrain department. The functions of this piece are to hold the bearings, the geartrain, the motor and the colling jacket. The cylindrical case hosts two bearings and the motor is held by eight threaded holes. In order to support the geartrain, three extrusions at the end of the body have been added. Furthermore, the extraction of the bearing is made possible by the two ears that have been modelled on the rib next to the body. Two junctions on the cylinder have been added in order to fix the inlet and the outlet of the cooling jacket.
- The upper left part, which connects to the a-arm of the suspension by a mounting and is connected to the kinematic support by thick plate, is later lightened. The face of the UBJ support and mounting have been angled to reduce the displacement relative to the ideal points of the kinematics in case of a setup changes on the camber, the default setting being -0.5° .
- The upper right part, that serve as a link for the tie rod. The same side has a ribbed plate, which serves as brake flange.
- The pushrod is connected through a mounting to a central support.
- The lower part connects to the inferior a-arm. A threaded hole has been added to fix the brake temperature sensor.
- Three cylindrical extrusions with threaded holes have been added on the main body to support the wheel cover.





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Bolts	They connect the mounting, the caliper and the tie rod to the upright, as well as the	Steel $E=210000 \text{ MPa}$	HEX (linear)

	mounting to the upper a-arm.	$\nu=0.3$	
Spheres	Part of the joint, allow a rotational degree of freedom to the a-arm.	Steel	HEX (linear)
Bushings	They distribute the preload of the bolts on a bigger surface	Steel	HEX (linear)
Dummy Geartrain	It transfers the torque to the upright	Steel	HEX (linear)
Dummy Hub	It simulates the brake disc and the hub with the races of the bearings	Aluminium (hub) Steel (internal races of bearings) Titanium $E=110000 \text{ MPa}$ (brake disc) $\nu=0.3$	HEX (linear)
Dummy Caliper	It simulates the brake caliper and the two bolts connecting it to the upright	Aluminium (caliper) Steel (bolts)	HEX (linear)
Seal protection ring	Sealing element for lubricants	Steel	HEX (linear)

Table 4-11 - properties

	Front Upright	Mounting	Bolts	Annulus Gear	Hub	Caliper
Boundary conditions	UBJ: hinged LBJ: X, Y translations locked S_U: Y-translation locked	Position locked by bolts	-	Position locked by bearings and hub	Disc tied to the brake pads	Position locked by bolts
Loads- Maximum Braking:	-3750 N (x) ground +270 (y) ground +2400 N (z) ground	-	13 kN preload (mounting) 10 kN preload (lbj)		2 kN preload on the hub	8 kN preload (bolts)

300 Nm(y), distributed on the 2 stages of the dummy geartain					
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Interaction properties

Surface to surface hard contact

Friction 0.3	Frictionless	Tie
Upright – Mounting	Dummy hub – Upright	Brake pad – Brake disk
Upright – Bolts		
Upright – Bearing races		
Upright - seal protection ring		
Mounting - Bolts		

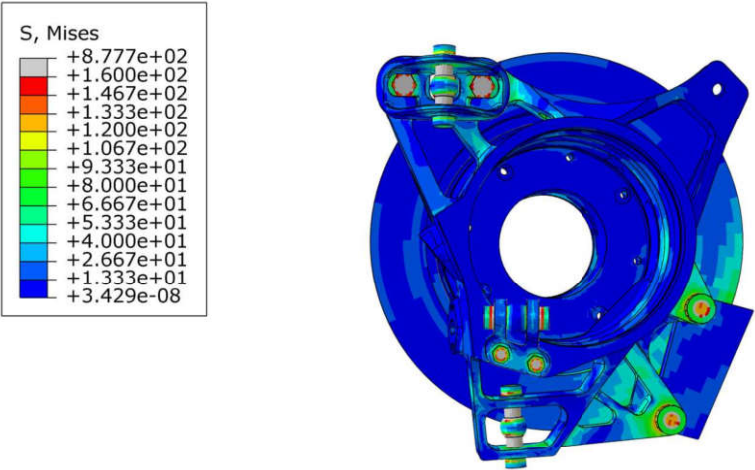


Figura 1RR

4.4 Rims

4.4.1 Goals

The main goals were:

- Improving stiffness-to-weight ratio
- Improving camber/toe compliances
- Allowing easier tire mounting

4.4.2 Constraints

The geometry has been constrained by:

- Overall dimensions and proportions of the AMK motor
- ETRTO Standard Manual
- Retro-compatibility with Keizer 10"x7.25" aluminium rims
- Brake disk and caliper position
- Draft angles to remove the part from the mould

4.4.3 3D Model

4.4.3.1 ETRTO Standard Manual Constraint

The channel shoulders geometry is based on the ETRTO Standards Manual 2017 (the European Tire and Rim Technical Organization). Taken the Hoosier ??/R10 tire as a starting point, the rim profile suggested by the manual has:

- Rim's width: 7 inches
- Rim's profile: 7j contour

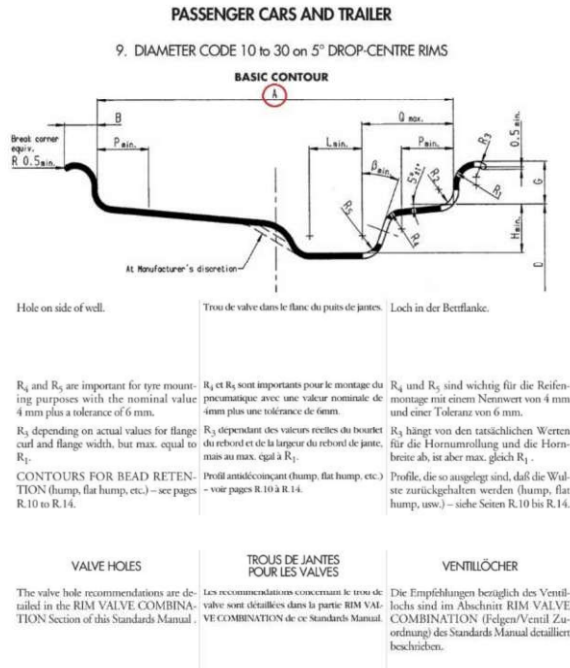


Figure 4-44 - ETRTO Standard

CONTOUR DIMENSIONS – CODE 10 to 30 on 5" DROP-CENTRE RIMS

RIM CONTOUR	DIMENSIONS (mm)											
	A	B	G	P	H	L	Q	R ₁	R ₂	β		
	min.	max.	± 0.6	min.	min.	min.	max.	min.	max.			
		(1)			(2)							
3.00B	76	± 1	10	13	14.1	13	15	16	28	7.5	4.5	10°
3.50B	89	± 1	10	13	14.1	15	15	19	34	7.5	4.5	13°
4.00B	101.5	± 1	10*	13	14.1	15	15	19	45	7.5	4.5	13°
4.50B	114.5	± 1	10	13	14.1	19.5	15	22	45	7.5	4.5	13°
5.00B	127	± 1	10	13	14.1	19.5	15	22	45	7.5	4.5	13°
5.50B	139.5	± 1	10	13	14.1	19.5	15	22	45	7.5	4.5	13°
6.00B	152.5	± 1	10	13	14.1	19.5	15	22	45	7.5	4.5	13°
3J	76	± 1	11	15	17.5	13	15	16	28	9.5	6.5	10°
3½J	89	± 1	11	15	17.5	15	15	19	34	9.5	6.5	20°
4J	101.5	± 1	11*	15	17.5	15	15	19	45	9.5	6.5	20°
4½J	114.5	± 1	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
5J	127	± 1	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
5½J	139.5	± 1	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
6J	152.5	± 1	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
6½J	165	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
7J	178	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
7½J	190.5	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
8J	203	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
8½J	216	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
9J	228.5	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
9½J	241.5	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
10J	254	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
10½J	266.5	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
11J	279.5	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
11½J	292	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
12J	305	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
12½J	317.5	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
13J	330	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
13½J	343	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°
14J	355.5	± 1.5	11	15	17.5	19.5	17.5	22	45	9.5	6.5	20°

(1) B max. values may be exceeded on rims for light commercial vehicles

(2) Minimum dimensions for well depth (H) and well angle are required for tyre mounting. For J Flange Rims of rim diameter codes 22 and above for new design, a well depth of 22 mm minimum is required.

Larger values may be required to ensure sufficient space for tubeless tyre valve seating.

For T Type tyres, rim code 15 in profile J and code 16 in profile B, might have a reduced flange width of 8 mm on the non-mounting side. The rim shall be marked with an additional "T" after rim width code.

Les valeurs de B max peuvent être dépassées pour les jantes des véhicules utilitaires légers.

Les valeurs minimales données pour la profondeur de gorge (H) et l'angle du flanc de gorge sont nécessaires pour le montage pneumatique. Pour les jantes à rebord J de jante code diamètre 22 et plus, pour les nouveaux modèles, une profondeur de gorge de 22mm minimum est exigée.

Des valeurs supérieures peuvent être nécessaires afin de ménager un espace suffisant pour l'assise de valve pour un pneumatique sans chambre.

Pour les pneumatiques de type T, le code jante 15 en profil J et code 16 en profil B pourraient avoir une largeur de rebord réduite de 8 mm sur le côté opposé au montage. La jante sera marquée d'un "T" supplémentaire, après le code largeur de jante.

Die Maße B max. können für LLKW-Felgen überschritten werden.

Die Mindestmaße für die Tiefe (H) und den Berflankenwinkel sind erforderlich für die Reifenmontage. Für neu entwickelte Felgen mit Hornausführung J und Durchmessercode 22 und darüber gilt für die minimale Tiefe des Felgenbetts ein Wert von 22mm.

Größere Maße können erforderlich sein, um ausreichend Platz für den Sitz des Ventils für schlauchlose Bereifung sicherzustellen.

J-Felgen für Reifen des Typs T mit Kode 15 sowie B-Felgen für Reifen des Typs T mit Kode 16 können auf der entgegengesetzten Seite der Montage eine reduzierte Felgenhornbreite von 8mm haben. Die Felgen müssen nach dem Felgenmaßkoden eine zusätzliche "T"-Markierung tragen.

See notes on pages R.4 to R.7.

DIAMETERS – CODE 10 to 30 on 5" DROP-CENTRE RIMS

NOMINAL DIAMETER CODE	10	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	28	30
SPECIFIED DIAMETER D (mm)	255.2	304.0	329.4	354.8	380.2	405.6	436.6	462.0	487.4	512.8	538.2	563.6	589.0	614.4	639.8	665.2	716.0	766.8

Figure 4-45 - ETRTO Standard

4.4.3.2 Keizer Retro-fitting constraint

From this sketch the actual channel profile has been designed. However, due to better retro-compatibility with Keizer rims, it's been chosen to modify "A" dimension, opting for a 184,15mm value (7.25 Inches) instead of 178mm.

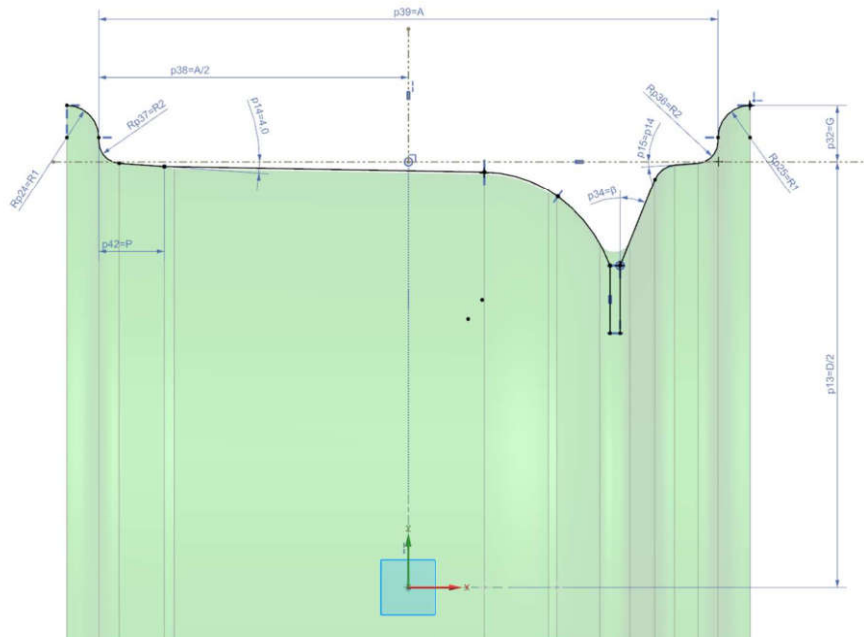


Figure 4-46 - Channel Profile Sketch